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COURSE NAME: CSA0905RA JAVA PROGRAMMING

CO3 AT3 – DEBUGGING

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, a 'Secure Exit' button, the lab title 'SIMATS Online Programming Lab', the subject 'JAVA PROGRAMMING', and a user profile for 'SARAN S' with ID '192473003RA'. The main area is divided into three sections: 'Your Fixed Code', 'Test Input', and 'Output'. The 'Your Fixed Code' section contains a Java program for a bank withdrawal simulation. The 'Test Input' section shows the input values '10000' and '3000'. The 'Output' section displays the program's output: 'withdrawal successful', 'Remaining balance =7000', and 'transaction completed'. A 'Pending manual review' status is shown at the bottom of the output section.

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[]args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             int balance=sc.nextInt();
7             int amount=sc.nextInt();
8             if(amount==0){
9                 throw new ArithmeticException("Withdrawal amount cannot be 0");
10            }
11            if(amount<0){
12                System.out.println("Invalid withdrawal amount");
13            }
14            else if(amount>balance){
15                System.out.println(" Insufficient balance");
16            }
17            else{
18                int remainingBalance=balance-amount;
19                System.out.println(" withdrawal successful");
20                System.out.println(" Remaining balance ="+remainingBalance);
21            }
22        }
23        catch(InputMismatchException e){
24            System.out.println(" Invalid input");
25        }
26        catch(ArithmeticException e){
27            System.out.println(" Transaction failed");
28        }
29        finally{
30            System.out.println(" transaction completed");
31        }
32    }
33 }
```

Test Input

10000
3000

Output

withdrawal successful
Remaining balance =7000
transaction completed

Pending manual review

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, a 'Secure Exit' button, the lab title 'SIMATS Online Programming Lab', the subject 'JAVA PROGRAMMING', and a user profile for 'SARAN S' with ID '192473003RA'. The main area is divided into three sections: 'Your Fixed Code', 'Test Input', and 'Output'. The 'Your Fixed Code' section contains a Java program for a patient registration simulation. The 'Test Input' section shows the input values 'Arun' and '25'. The 'Output' section displays the program's output: 'Patient Name :ARUN', 'Adult patient', and 'Registration completed'. A 'Pending manual review' status is shown at the bottom of the output section.

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[]args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             String inputName=sc.nextLine();
7             String name;
8             if(inputName.equals("NULL")){
9                 name=null;
10            }
11            else{
12                name=inputName;
13            }
14            int age=sc.nextInt();
15            if(name==null){
16                System.out.println("Patient name is unavailable");
17            }
18            else{
19                System.out.println("Patient Name :"+name.toUpperCase());
20            }
21            if(age>=18){
22                System.out.println("Adult patient");
23            }
24            else{
25                System.out.println("Minor patient");
26            }
27        }
28        catch(InputMismatchException e){
29            System.out.println(" invalid age input ");
30        }
31        catch(Exception e){
32            System.out.println("Registration failed");
33        }
34        finally{
35            System.out.println("Registration completed");
36        }
37    }
38 }
```

Test Input

Arun
25

Output

Patient Name :ARUN
Adult patient
Registration completed

Pending manual review



Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

SARAN S

192473003RA

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String[] products={"Laptop","Mobile","Tablet","Headphone","Keyboard"};
6         try{
7             int productNumber=sc.nextInt();
8             if(productNumber<1||productNumber>5){
9                 System.out.println("Invalid product number");
10            }else{
11                System.out.println("Selected product :"+products[productNumber-1]);
12            }
13        }catch(InputMismatchException e){
14            System.out.println(" Invalid input");
15        }catch(Exception e){
16            System.out.println("Search failed");
17        }finally{
18            System.out.println("Search completed");
19        }
20    }
21 }
22
23
24

```

Submitted

Test Input

1

Output

Selected product :Laptop

Search completed

Pending manual review



Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

SARAN S

192473003RA

Your Fixed Code

```

1 public class Main{
2     public static void main(String[] args)throws InterruptedException{
3         BankAccount account=new BankAccount();
4         ATM atm1=new ATM(account,7000);
5         ATM atm2=new ATM(account,6000);
6         atm1.setName("ATM-1");
7         atm2.setName("ATM-2");
8         atm1.start();
9         atm2.start();
10        atm1.join();
11        atm2.join();
12        System.out.println("Final Balance = "+account.balance);
13    }
14 }
15 class BankAccount{
16     int balance=10000;
17     synchronized void withdraw(int amount){
18         if(balance>=amount){
19             System.out.println(Thread.currentThread().getName()+" withdr
20         }
21         try{
22             Thread.sleep(100);
23         }
24         catch(InterruptedException e){
25             System.out.println("Interrupted");
26         }
27         balance=balance-amount;
28         System.out.println(Thread.currentThread().getName()+" withdr
29     }else{
30         System.out.println(Thread.currentThread().getName()+" insu
31     }
32 }
33 class ATM extends Thread{

```

Test Input

stdin input (optional)

Output

ATM-1 withdrawing 7000

ATM-1 withdrawal succesful

ATM-2 insufficient balance

Final Balance = 3000

Pending manual review

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Reservation reservation=new Reservation();
5         Customer c1=new Customer(reservation,"customer 1");
6         Customer c2=new Customer(reservation,"Customer 2");
7         c1.start();
8         c2.start();
9     }
10 }
11 class Reservation{
12     int seats=1;
13     synchronized void bookSeat(String customer){
14         if(seats>0){
15             System.out.println(customer+" is booking a seat");
16             try{
17                 Thread.sleep(100);
18             }
19             catch(InterruptedException e){
20                 System.out.println("Thread interrupted");
21             }
22             seats--;
23             System.out.println(customer+" booking successful");
24         }else{
25             System.out.println(customer+" - no seat available");
26         }
27     }
28 }
29 class Customer extends Thread{
30     Reservation reservation;
```

Test Input

stdin input (optional)

Output

```
null is booking a seat
null booking successful
null - no seat available
```

⚠ Pending manual review